NightSweep 1.8 Alpha build 2 — All Functions Reference

Technician README · Updated November 11, 2026

This reference explains the working functions in NightSweep 1.8 Alpha build 2: where to find them, what to enter, how to interpret the result, and which bundled example lets you practice. It covers all 20 task screens and the 37 entries in the Practice library's feature index.

For installation, use the separate Android and iOS installation guide. This reference applies to the shared Android/browser application. Browser-only offline and update controls appear on Home; Android's native save/share/print dialogs appear when using the APK.

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Working rules and navigation

Start a real job with **Job → Details → + New blank job**. Identify the actual source output, enter launch readings, build the route, then use the calculations, setup guides and records you need. A single-span calculation does not require completion of the advanced acceptance forms.

Main work area	Task screens	Purpose
Home	Home	Current job, task shortcuts, practice, backup and browser offline status
Route	Setup; Cable route; Amp gain; Equipment setup; Equipment	Starting signal, RF topology, active gain and exact-model references
Test	Sweep setup; TDR; Sweep; Frequency; Readings; References	Guided test records, predicted response, distance and measurement comparisons
Power	Load; What-if	Independently entered power circuit and an added-load trial
Job	Details; ONX files; Checklists; Records; Run sheet	Local jobs, original evidence, work notes, acceptance records and reports
Header / Home	Help & methods	Task help, methods, limits and build self-check

The small task tabs choose a function within the main area. The point selector and **Previous/Next** choose the route location; they do not change its feed parent. **More at this point** exposes tap faces, drop, map notes and arithmetic.

Keep these distinctions while using every tool:

- **Prediction** is calculated from entered data. A meter reading is a separate entry; neither automatically becomes acceptance.

- **Manufacturer capability**, local design target, installed configuration and available spectrum are separate fields.
- A dash or blank result means unavailable, missing, invalid or outside supported data. It is not zero. A deliberately entered zero is different.
- Use the same frequency, measurement plane and signal-power convention when comparing readings. Apply only test-point correction the meter has not already applied.
- **Vecima nodes / AOI actives:** local 204/258 MHz workflow, forward Sweepless Sweep and XPERTrak return sweep. **GS7K, GainMaker and the other fleet families:** local 42/54 MHz workflow, forward and Return Sweepless Sweep.
- Non-high-split design anchors are **31 dBmV at 54 MHz / 43 dBmV at 860 MHz**. The separate **38 dBmV return TX reference** does not specify a receiver target, injection setting, channel or measurement plane.

Header and Home controls

Where: the header and Home. **Practice:** Complete field workflow; Custom curves for Help and methods.

Function	Enter or do	Result and limit
Main and task tabs	Choose a work area, then its task	Opens the relevant controls while preserving the current job
Files → Save project	Export the working project and finish the device's save action	Portable JSON with supported job data and original evidence; verify the file outside the app
Files → Open project	Choose a NightSweep project JSON	Validates the incoming project before loading; export important current work before replacement
Files → Export RF levels	Export the current route table	Predictions and context in CSV; unavailable results remain blank
Files → Print report	Open the native or browser print dialog	Printing/PDF depends on the device and print service; use Run sheet for a focused field summary
Help & methods	Open help from the header or Home	Task guidance, calculation definitions and limits
Sun / Night	Select the display palette	Changes presentation, without changing inputs or calculations
Working-job card	Read the job name, asset and sample/training warning	Helps prevent entering real readings into a sample job
Task shortcuts	Choose route, TDR, gain, readings, power, sweep setup or run sheet	Opens that live task using the current job

Function	Enter or do	Result and limit
Home statistics	Review farthest route distance and active count	Summarizes the entered RF topology; an active count alone does not prove a serial cascade
New-user quick start	Expand the short workflow	Opens existing task controls; it does not complete them automatically
Home backup / project opener	Use the Save/Open project shortcuts	Same full-project workflow as Files; not automatic cloud synchronization
Status and warning messages	Read the nearby message after an action fails	Identifies missing inputs, storage/file errors or a blocked operation; preserve backups before repairing storage

Browser Home also provides **Install app**, **Check for updates**, **Install update**, **Reload app**, **Retry offline setup** and **Protect local storage** when supported. Their behavior is explained under Offline use, updates and Help.

Practice library and example selection

Where: Home → Practice library. Choose an exercise, or choose a feature to load its seeded example and open the matching guided step. These examples are bundled; no manufacturer website is needed to load them.

Exercise name	What is loaded	Guide / supplied evidence
Complete field workflow	GS7K source, two GainMaker GaN HGD actives, Scientific Atlanta / GigaXtend / Technetix passives, branched 42/54 route, power, readings and records	21 steps; five synthetic CSV originals and one original-only SVG diagram
Six active devices in series	Vecima EN9000 source and six serial AOI actives on 204/258	13 steps; no supplied meter captures
Custom curves and reverse TDR	Manual launch table, custom cable/passive tables, exact-frequency entries and reverse TDR	8 steps; no supplied meter captures
GainMaker OEM AGC	Separate 42/54 conventional OEM detector/pilot path	8 steps; no supplied meter captures
GainMaker / DSIM	Separate 42/54 fitted DSIM identification, historical code lookup and incomplete verification	9 steps; no supplied meter captures
GainMaker manual / thermal	Separate 42/54 manual/thermal installed-control path	7 steps; no supplied meter captures

Exercise name	What is loaded	Guide / supplied evidence
GS7K conventional node	Standard GS7000 1.2 GHz / 42/54 setup starter; actual launch/readings/correction unverified	10 steps; no RF route or supplied meter captures

Opening or resetting training preserves the working job locally before switching. A full or unreadable named-job list, an evidence operation in progress, or a preservation failure stops the switch. Read the message and keep the existing job; do not clear storage to force it. A local retained copy is not an external backup.

Use **Open step** to reveal the relevant controls, try them, then mark **Reviewed**. **Practice guide** in the training banner returns to the guide. Progress is saved with that practice copy; it does not mark field inspections or acceptance complete. Reset preserves the edited copy before restoring the exercise baseline.

Older practice copies remain readable. If their equipment no longer matches the current complete exercise, the app hides mismatched guidance and offers an explicit reset. Renaming, editing, saving or exporting a practice job does not remove its synthetic identity.

The complete workflow uses GS7K/GainMaker GaN, not the AOI six-active path. Its GS7K hardware capability is 1218 MHz, but the entered operating/planning upper is **1000 MHz** and live spectrum **860 MHz**. The selected GaN active covers 54–1002 MHz and the Scientific Atlanta route passive stops at 1000 MHz. Do not infer end-to-end 1218 MHz operation from the source model.

Job details and named jobs

Where: Job → Details. **Practice:** Complete → Job details; any exercise's backup step.

Function	Inputs and use	Output and limit
Job identity	Name, node/asset, date and lead technician	Current-job context used in reports and exports
Work-order fields	Ticket, address/location note, symptom, start/end times, weather note, resolution and parts used	Stored with the job; a time-order warning preserves the entered text and does not change calculations
Linked metadata	Edit work-order number, node or lead in Details or Records	Keeps those shared identity fields consistent; other record fields remain their own entries
Node details	Identify model, installed band, asset, RF port/leg, modules, operating upper and notes	Describes source identity; model selection does not supply measured launch levels

Function	Inputs and use	Output and limit
+ New blank job	Start a clean field job after protecting current work	Removes sample inputs from the new working job; check its name before field entry
Save named job	Save the desired working version to the device's job library	Local named copy; save/export important revisions deliberately
Open	Select a stored job	Restores that job's data and attachment references; storage failure blocks unsafe switching
Duplicate	Make a separately named copy	Uses the latest working state when duplicating the active named job; updated original and new copy are written together. A failed write leaves the library unchanged; training stays synthetic
Delete	Remove the intended named entry after exporting anything needed	Not a recovery workflow; check the job identity before confirming

Autosave and named jobs use this app/browser's storage. APK, Chrome, another browser/profile, another phone and a Home Screen web app may use different stores. No account or automatic job synchronization is provided. Use one NightSweep window at a time and transfer jobs with verified full-project exports.

RF setup and launch source

Where: Route → Setup. **Practice:** Complete → Launch; GS7K conventional node → Identity; Custom curves → Launch.

Function	Inputs and use	Result and limit
Launch-source choice	Select **Node output**, **Amplifier / LE output**, or **Manual / other output**	Identifies the physical starting output; it does not add an artificial gain block or fill levels from a rating
Node / active identity and output	Match exact model, installed split/module and selected RF port; use the job node ID shortcut where appropriate	Makes the source/manual context explicit; source changes preserve readings for review rather than making them valid for the new connection
Source setup & manual	Open the selected equipment's reference	Requires the matching installed variant and revision; external originals need internet
Low/high launch markers	Enter two frequencies and corresponding levels at the same source plane	Defines the two-point source curve; ordered, valid in-band markers are required

Function	Inputs and use	Result and limit
Launch test point	Mark raw TP readings and enter only remaining coupling	Adds the stated correction once; use zero/off when the meter already supplies mainline-equivalent levels
Launch curve	Choose square-root frequency, linear MHz or **My level table**	Uses the selected interpolation model; it is not a measured EQ response
My level table	Enter one `MHz, dBmV` pair per line	Uses actual entered source points; stops outside the first/last supported point
Extend two-point launch slope	Enable only for an intentional estimate outside the two anchors	Extends the two-point launch estimate; it does not extend cable/passive table coverage or establish live carriers
Available spectrum ends	Enter the known live-spectrum upper limit	Distinguishes live from planning predictions; hardware capability is separate
Passive loss basis	Choose **Nominal / typical** or **Maximum**	Nominal can use a labelled maximum fallback when no nominal curve exists; read the stated basis
Cable temperature	Enter RF cable temperature in °F	Adjusts supported built-in attenuation references; custom cable values are already at actual temperature
Next: cable route	Continue to the route editor	Starts at the chosen source output without inventing another station

In Complete, **11/23 dBmV +20 dB** at **54/860 MHz** produces **31/43 dBmV** launch. This is synthetic arithmetic, not proof those physical carriers or readings are present. The six-active AOI example instead uses 261/993 MHz and 34/46 dBmV.

Cable route, branches and point tools

Where: Route → Cable route. **Practice:** Complete → Route / Drop; Custom curves → Custom / Arithmetic; Six actives → Route.

Function	Inputs and use	Result and limit
Add, select and remove points	Add a cable span ending at equipment; select it with the point selector or Previous/Next	One point includes its incoming cable. Review child feeds after changing/removing a point; maximum 200 route points
+ Append after last tap	Use when the final route entry is a tap with a through output and a lower published face available	Appends through that tap, copying cable, incoming length, slack and custom cable data while choosing the next lower published face. It does not choose an arbitrary earlier branch or work after an ineligible final point

Function	Inputs and use	Result and limit
Point name and incoming feed	Name the location and choose the actual parent **output**	Builds the topology. A branch feeds from its selected splitter/tap port, not merely the same housing
Cable, length and slack	Select cable, enter mapped feet and additional slack	Adds both to attenuation/distance; do not add prefilled slack again
Apply this length to following spans	Confirm copying the selected point's incoming length	Applies to later entries in the route list, including later branches; not only the chosen electrical path. Review affected points and their separate slack
Equipment selection	Choose the exact family, type, part/face and terminal configuration	Uses the selected port curves; active, passive, cable-only and custom points have different behavior
Parent outputs and branch ports	Choose through, low-loss, high-loss, tap or other actual output	Preserves unequal splitter paths. Per-F-port tap output is not the through level
Cable-only point	Select no ending passive	Retains the cable span without adding invented equipment loss
Custom cable	Enter `MHz, dB per 100 ft` rows	Uses entered attenuation at actual temperature; no extra built-in temperature factor or out-of-range extrapolation
Custom passive	Enter `MHz, through dB, tap/branch dB` rows	Keeps the two output losses separate; missing data remains unavailable
Published faces	Compare same-family / same-port-count tap faces at the chosen frequency; choose a face to swap the part	Changes the catalog part and prediction; read through and tap loss separately
RG-6 drop off a tap port	Plan a drop from a selected tap output; enter feet, a custom cable table, then no splitter, a supported splitter or explicit custom loss	Calculates drop-end outputs independently of the mainline route; this is not a power-passing model
Remove drop	Remove the optional drop plan	Does not remove the mainline tap
As-built / map variance	Record a location-specific note	Saved with the point and shown in relevant reports; it is not a calculated cable measurement

Function	Inputs and use	Result and limit
Show the arithmetic	Expand the selected point's detailed calculation	Shows source/input, cable rate/temperature/feet, equipment loss or gain and output; unavailable steps stay visible as unavailable
Route cascade / distance summary	Follow the selected feed path and branch totals; scroll within the compact cascade in either direction, or focus its controls by keyboard	Shows separate path coordinates; all points remain reachable without making the phone page excessively tall. Branch footage is not added to the main endpoint
3-tap route example	Load the small RF-route sample after protecting work	A separate, simpler example; it is not the complete training job or the six-active run

Complete uses Scientific Atlanta **SAT MM 4-23**, GigaXtend **XSC-2-959** and **GTSG-TAP-4-20-STD**, and Technetix **OTTZ-4-17**. Its drop uses a **75 ft synthetic RG-6 table plus explicit custom 4 dB loss**, not an asserted premises-splitter model. Its main endpoint is 2380 ft from source; the branch cable-only endpoint is 2220 ft.

Amplifier gain and test-point helper

Where: Route → Amp gain. **Practice:** Complete → Gain; Six actives → Gain / Gain table.

Function	Inputs and use	Output and limit
Active-point selector	Choose an active already in the RF route	Edits that station; source identity alone does not create a route amplifier
Two measured/configured gain points	Enter the two frequencies and available operating gain at each	Calculates frequency-specific gain; match the installed configuration and measurement planes
My frequency/gain table	Enter `MHz, operating gain dB` rows	Uses the entered response within its supported domain
Additional selected-output loss	Enter only loss after the gain's measured output plane	Subtracts that extra loss; leave zero when the splitter/pad is already included in the measured net gain
Desired low/high output	Enter the station's required targets, or review defaults from corrected launch	Determines gain needed after the preceding cable/passives; a target is not measured output
Desired gain reserve	Enter the requested reserve in dB	Separates meeting a target from retaining adequate margin
Optional installed output limits	Enter verified limits for model, channel loading, tilt and output path	Checks requested/predicted output against those entries; blank means capability unchecked

Function	Inputs and use	Output and limit
Gain budget	Review required gain, entered gain, low/high margins and reserve status	Can show missing, invalid, shortage or marginal reserve. One good marker does not compensate for another bad marker
Manufacturer gain reference	Read the source, frequency/configuration and qualification	Screening reference only; a scalar rating does not become a flat gain curve or prove maximum available gain
Calculate gain from test-point readings	Enter input/output readings at the same frequency plus their separate remaining corrections	Corrected output minus corrected input; no gain is inferred from an output-only photo
Use as low / high gain point	Apply the helper result to the chosen gain marker	Updates the calculation input; does not align physical hardware

Complete's helper uses **7.1/16.1 dBmV +20 dB each** at 54 MHz: corrected 27.1/36.1 dBmV gives **9 dB gain**. A1's entered anchors are 9/18 dB at 54/860 MHz; A2's table is 54 MHz/8 dB, 550 MHz/12 dB, 860 MHz/15 dB. These are synthetic net gains, not GainMaker GaN control settings.

Installed equipment, manuals and control paths

Where: Route → Equipment setup; also **Source setup & manual** from Setup. **Practice:** Complete; OEM AGC; DSIM; Manual/thermal; GS7K node; Six actives.

Function	Inputs and use	Output and limit
Equipment to set up	Select the launch node/active or a route active	Opens that exact station's saved draft and applicable reference context
Identity and manual review	Match model/module, split, port, firmware, manual revision and measurement plane	Displays source-qualified steps and references; a product brief is not a commissioning procedure
Setup frequencies & job levels	Read selected-profile base anchors, actual-carrier draft and measurement-plane reminders	Keeps design targets, estimated carrier targets, approved pilot and procedure backoff frequencies separate
Installed gain control	Select **OEM GainMaker AGC**, **ACI DSIM**, or **Manual / thermal control** where applicable	Selects the matching guide; changing hardware/control/pilot context can invalidate prior reviewed steps
Documented setup method	Choose the exact station's applicable manual-backoff or thermal branch	Supplies source-specific guidance. Never combine branches/revisions because a setting looks familiar
Reviewed steps and station notes	Record what was reviewed and unresolved	Persists with that station; not proof of work, pilot lock or RF acceptance

Function	Inputs and use	Output and limit
External source links	Open the original or explicitly labelled historical reference	Needs connectivity; verify exact installed applicability and document revision

Conventional GS7K: the node guide distinguishes receiver/FCM/RCM/routing, selected-port pads/EQs and the actual reverse transmitter/EDR/RPD. Conventional LCM Auto Set-Up discovers modules; it does not balance RF. Intelligent, SHO/Fiber Deep and retrofit-RPD references require their own hardware/procedure and are not interchangeable conventional-node profiles.

Complete's GainMaker GaN HGD: exact installed GaN commissioning/control instructions remain unresolved. The app keeps this gap visible. Conventional GaAs OEM pad/backoff formulas, ACI DSIM instructions and AOI alignment sequences do not establish GaN settings.

Focused OEM AGC: match the detector, approved pilot and manual. The displayed manual-backoff and thermal preparation branches are alternatives. Conventional GaAs system-amplifier and LE references use different pad relationships and different backoff tables; the station/manual must match. No universal OEM QAM detector offset or settling timer is supplied.

Focused manual / thermal: identify the installed control path and follow its own applicable method. Do not infer OEM AGC pilot-pad values or DSIM codes from that selection.

ACI DSIM functions:

Function	Inputs and use	Result and limit
Module / firmware / approved pilot	Identify DSIM variant, controller/module revision, pilot MHz/type and approved operating context	Keeps the setup bound to the installed equipment; DSIM-SG and later DSIM-A variants are not assumed equivalent
Physical-controller/app guidance	Read the exact cited interface/procedure and temperature preparation	Source-specific review only; physical RF adjustments and raw knob counts are not generic app dB settings
Historical pilot lookup	Choose channel or exact frequency and Analog/Digital type	Returns the literal ACI Rev A code when available. Channel 110 digital is 711 MHz, `@@?<A`
Copy code / manual fallback	Copy from a supported user action, or select the code for manual copying	No device is programmed; copying does not verify compatibility, a supervisor key or pilot lock
Missing/anomalous lookup rows	Read the unavailable/anomaly message	No codes are invented for 95–99, 303 MHz or OFDM PLC; channel 100 copying is blocked as a source anomaly
Key/review status	Record the review flag supported by the form	Do not enter or share an actual supervisor key; a flag is not credential verification

The historical Rev A table contains 86 literal rows: channels 52–94 and 100–142. DSIM-SG Rev E's pad instructions differ from Rev H; the app preserves that conflict rather than mixing settings. Match the full installed procedure before physical use.

AOI in the separate six-active example: HGD/HGBT guidance covers saved downstream profile → downstream Auto → optional downstream Manual → applicable upstream sequence → ALSC after both paths. The cited historical SA procedure is not a substitute for missing exact Quantum18 LE/BA I&O. Its instrument instructions remain separate from the local XPERTrak return workflow.

Equipment catalog

Where: Route → Equipment. **Practice:** Complete → Equipment references.

Choose category/family and search the exact model. The list includes RF equipment, cables, node profiles and power models. Search can widen beyond a family when that filter has no matching entry; read the displayed count/filter message and narrow a long list.

Expand **View published loss data**, **View operating gain & conditions**, **View bands & test points**, or **View power data & conditions**. Read units, frequency/voltage domains, source, configuration, nominal/maximum basis and source qualifications before using a value. Node and power entries link to their working setup/load screens.

Custom entries require actual input tables. Published references do not automatically fill installed operating gain. Unavailable curves do not become zero or extrapolate. UBT remains unavailable for new selections; older saved UBT definitions remain readable without silently replacing them. A legacy power row with unsupported current data needs an applicable entered load.

TDR distance locator

Where: Test → TDR. **Practice:** Complete → TDR; Custom curves → reverse TDR; Six actives → reconnect after A6.

Function	Inputs and use	Result and limit
Route endpoint/path	Select the actual endpoint whose feed path is tested	Excludes unrelated sibling branches
Connection point	Select Launch or the correct cable side after a route point	Changes the origin; route coordinates remain measured from the source
Direction	Choose toward endpoint or toward source	Supports forward and reverse testing along the selected path

Function	Inputs and use	Result and limit
Displayed distance and lead	Enter the meter's one-way displayed distance and physical test lead	Corrects the path distance; do not enter a round-trip time as feet
VF correction	Enter meter-set and actual velocity factors, enabling correction when needed	Scales displayed distance by actual VF / meter VF, then accounts for the physical lead
Locate result	Review corrected path distance, route coordinate, span and offsets	Places an entered distance on entered geometry; does not classify or detect a physical fault
Active boundary / invalid input message	Read the first barrier or missing/invalid condition	Uses the actual parent-chain order, including distinct active points sharing one route coordinate. A TDR path cannot cross another amplifier; reconnect on the correct cable side

Complete: after GaN A2 at **1700 ft**, **295 − 25 = 270 ft** of cable path locates **1970 ft**, **160 ft** into Main Tap B's 310 ft span. Both VFs are 0.87. Launch-origin testing beyond the active is blocked.

Custom curves: reverse from the 200 ft endpoint; **100 × 0.88/0.80 − 25 = 85 ft** toward the source places the hit at **115 ft**, 15 ft into the second span. These VFs and distances are teaching inputs.

Forward and return sweep setup

Where: Test → Sweep setup. **Practice:** Complete for 42/54; Six actives for 204/258; focused equipment exercises for their matching procedures.

Function	Inputs and use	Result and limit
Split/workflow selection	Select 42/54 or 204/258 for the actual fleet/configuration	Retains separate setup drafts; changing a tab does not reconfigure hardware
Plant, equipment and reference	Record station/leg, equipment, instrument/firmware/options, service/sweep plan, reference, available band and measurement planes	Makes the comparison reproducible; instrument capability and compatible references still require verification
Guided setup	Use Previous/Next step and reviewed checkboxes; record actual settings/findings	Saved guidance/progress, not remote control or acceptance
42 MHz return sweepless	Use the approved DOCSIS-derived referenced response, actual service and matching saved reference	Comparison is in dB; no absolute physical receiver level or generator setting is inferred

Function	Inputs and use	Result and limit
Operator return TX reference	Record 38 dBmV separately, plus actual TX frequencies/service plan and transmitter/plane	A local level reference supplies no missing channel, return-injection or receiver target
Compare return markers	Enter actual low/high frequencies, readings, targets and tolerances in matching units	Reports differences and tilt against entered references; check the rest of the trace separately
204 MHz comparison basis	Choose referenced response (dB) or a verified absolute receive plane (dBmV)	Relative and absolute values are not interchangeable
Return sweep injection budget	For the 204 workflow, enter desired network-plane injection, total physical coupling/lead losses and applicable generator maximum	Estimates physical connector output and checks the entered limit; it is not automatically the ONX Injection field setting
Closeout notes and sources	Record findings, gaps and trace references; review original source links	No ONX/XPERTrak connection or equipment programming is performed
Save station snapshot	Save the station draft before editing another station	Adds a labelled snapshot to history, up to 100; preserves notes/readings, not acceptance
Saved station snapshots	Review history with station, split, time, leg/reference and training identity	Export includes snapshots and both current split drafts; it is not an external backup by itself
Export setup notes	Finish the text-file save/share action	Exports the entered setup, guidance and history; keep originals/full project separately

For physical-output budgeting, full physical losses remain losses even if instrument compensation is enabled. Do not enter the summed physical-output estimate as ONX network-plane Injection while also applying the same TPC. The Records injection helper instead uses **remaining uncompensated** losses for its manual-setting estimate. Neither establishes an approved meter mode or measured injection. Return Sweepless Sweep does not use this generator calculation.

Predicted sweep and output table

Where: Test → Sweep. **Practice:** Complete → Predicted sweep; Six actives → six restored marker outputs.

Function	Inputs and use	Result and limit
Trace selection	Select source and relevant route outputs	Shows the selected paths; compare through and tap ports separately

Function	Inputs and use	Result and limit
Signal level	Choose absolute dBmV mode	Predicted output versus frequency, dependent on valid source and component data
Normalized path response	Choose dB mode and reference MHz	Removes the launch shape and zeros the supported path response at that reference
Total path loss	Choose loss mode	Net path loss, including active effects where supported; not a measured sweep trace
View range and reference	Enter view start/end and supported normalization frequency	Changes the chart view/reference; does not change equipment capability or available spectrum
Pointer/touch readout and legend	Tap/move over the chart and read trace labels	Frequency/level readout for calculated data
Live/planning display	Review shading, solid/dashed presentation, gaps and chart notes	Above-live values are planning predictions. Gaps may be missing tables or launch support; never treat them as zero
Signal at each tap/output	Review the tabulated frequencies, ports and tilt/context	Tap results are at the F-port before the optional drop; fixed table markers can be unavailable on this route

Launch slope extension, when enabled, remains an estimate. Cable/passive coverage is not extended. Build 2 defers expensive chart redraw while this screen is hidden and draws current data when it is shown or printed. Frequency results are reused only within one synchronous refresh, not across later edits or imports. The model does not reproduce narrowband ripple, ingress, micro-reflections or a measured fault signature. A 993 MHz result cannot establish 1.2/1.8 GHz plant performance.

Single-path frequency checker

Where: Test → Frequency. **Practice:** Complete → Frequency; Custom curves → exact overrides.

Function	Inputs and use	Result and limit
Cable/passive combination	Select cable, feet, passive, actual path and quantity	A separate combination; does not add route points or alter RF topology
Exact frequency input	Enter a supported frequency from 5–1800 MHz	Returns the calculation and band/coverage flags at that point
Exact cable/passive overrides	Supply known dB/100 ft and loss-per-device at that frequency	Uses entered values already at stated conditions; clearing them falls back only to available reference data
Optional launch at that frequency	Enter a level at the matching plane	Computes receive budget after total loss; blank launch leaves receive level unavailable

Function	Inputs and use	Result and limit
Equipment/spectrum/ passband limits	Enter installed, planned/live upper limits and split	Shows conflicting/out-of-band conditions; broad software range is not equipment support
Frequency table	Inspect low/high and other listed markers	Shows cable rate/loss, passive each/total, total loss, launch, receive and basis/status
Complete route / span summary	Select an existing route output	Summarizes that whole feed path and distance; different from the single combination above
Export frequency results	Export CSV and finish saving	Includes inputs, units, flags and unavailable values

Complete at **860 MHz**: **200 ft ×2.2 dB/100 ft + 1.4 dB =5.8 dB** total; **43 − 5.8 = 37.2 dBmV** receive. Custom curves at 261 MHz: 100 ft ×1 dB/100 ft +1 dB =2 dB, leaving 32 dBmV from 34 dBmV. These are exact-frequency teaching inputs.

Meter readings and prediction error

Where: Test → Readings. **Practice:** Complete → Meter readings; Six actives → A6 comparison.

Choose a route point and actual output, then enter frequency, reading, test-point status and remaining coupling. The comparison shows raw reading, corrected reading, prediction and **measurement minus prediction**. A negative error means the entered measurement is below the model.

Use zero/off for remaining correction when the meter already reports mainline-equivalent level. A reading at another frequency is not a valid comparison with the chosen marker. Missing predictions/readings remain unavailable. This task does not automatically fill the official forward/return records or approve their criteria.

Complete includes intentional mismatches. For example, A1's entered **16.1 + 20 = 36.1 dBmV** at 54 MHz is compared with about **36.94 dBmV** predicted. Tap C's direct **7.50 dBmV** compares with about **14.01 dBmV**. Investigate inputs and measurement planes; synthetic disagreement is not evidence of a physical defect.

Field references and target conversions

Where: Test → References. **Practice:** Complete → References; DSIM/OEM/GS7K → available carriers.

Function	Inputs and use	Result and limit
Your local non-high-split targets	Apply the labelled 31 @ 54 / 43 @ 860 · TX 38 preset to a compatible selected profile after confirmation	Updates its stated design/reference fields; does not write measured launch, approve criteria or supply missing measurements

Function	Inputs and use	Result and limit
Preset compatibility checks	Match hardware and forward passband containing both base anchors	Blocks high-split/incompatible use; a low return ceiling alone does not prove 54 MHz is usable
Setup targets at available carriers	Enter actual low/high channel centers and choose square-root frequency or linear MHz	Estimates targets from the original 54/860 anchors; range must satisfy 54≤low<high≤860 and known live/hardware limits
Fill 57/753 or 63/550 example	Fill only the converter draft	Does not replace base design anchors, measured launch or approved pilot
Common launch anchor pairs	Review/apply the labelled frequency preset	Changes its stated fields only; an example is not a live carrier survey
Editable node/launch target ranges	Use the selected profile or labelled workbook/sample draft; edit min/max/tolerance	Creates draft targets. **Apply ranges to selected profile** changes that selected profile, keeping approval separate
Published tap faces quick look	Select the route tap and frequency; open its face picker	Compares compatible published parts; choosing a face changes that route part
Cable loss quick look	Choose cable and frequency	Reference loss with source/domain qualifications; not an entered span measurement

Square-root examples from 31/43 dBmV at 54/860 MHz: **57/753 → 31.11/41.97 dBmV**; **63/550 → 31.32/39.79 dBmV**. The corresponding linear-MHz estimates are **31.04/41.41** and **31.13/38.38**. Curve choice requires the applicable local procedure; these are not a measured installed equalizer curve.

The available-carrier draft survives project backup and is reported separately in equipment setup/run sheet. It does not choose an AGC pilot. If actual live spectrum stops below the practice pilot, identify a compatible approved live pilot; the app does not invent one. The six-active Vecima/AOI profile keeps its separate in-band 261/993 MHz setup.

Power-supply load

Where: Power → Load. **Practice:** Complete → Power; Six actives → explicit power topology.

Function	Inputs and use	Result and limit
Supply voltage/rating	Enter coax output V RMS and A RMS rating	Defines the supply-side calculation and loading/headroom; not utility-mains current

Function	Inputs and use	Result and limit
Add equipment / junction	Enter each powered device once; use a zero-load junction for branches	Builds the actual power circuit independently of RF route parents
Power parent	Choose supply or the actual upstream power point	Shared sections carry downstream branch currents; RF direction alone does not establish this
Cable and resistance	Enter feet, supported cable or custom loop resistance and extra/contact resistance	Calculates section voltage drop and resistive loss. Custom resistance is used as entered at actual temperature
Power-cable temperature/coefficient	Enter the power temperature and applicable resistance coefficient	Adjusts supported nominal loop resistance; separate from RF cable temperature
Manufacturer current curve	Select the exact supported configuration	Interpolates current within its documented voltage domain; no borrowed curve or out-of-domain extrapolation
Entered amps	Enter known/measured current	Uses that load directly; synthetic examples are not manufacturer specifications
Entered input watts / PF	Enter watts and valid power factor	Estimates current from equipment voltage and PF; unsuitable/inconsistent inputs prevent a trustworthy result
Entered input VA	Enter apparent power in VA	Estimates current as VA divided by equipment voltage; power factor is not applied a second time
Minimum equipment voltage	Enter the applicable installed limit if known	Supports voltage warnings; a curve's table boundary is not automatically an undervoltage cutoff
Supply summary	Read total output current, loading %, headroom, wire loss and minimum equipment voltage	Scalar RMS planning result, not a measured quasi-square-wave waveform
Per-section table	Read equipment V, device A, section A, section drop V and loss W	Distinguishes a device's current from all current carried through its incoming section
Remove / edit row	Change only the intended circuit point and inspect child feeds	Verify actual power passing, supply boundaries and connector ratings independently
Export power load	Save the power CSV	Includes circuit inputs, configuration/source and warnings; not a full-project backup

Complete uses a **90 V / 15 A** supply; synthetic entered **0.75 A GS7K**, **0.60/0.45 A GaN** loads and a **20 W / PF 0.9** branch. Its independent reference result is **2.047 A**, **13.65%** loading and **88.83 V** minimum at GaN A2. The GaN selection has no verified current curve; no AOI curve is substituted.

Power what-if

Where: Power → What-if. **Practice:** Complete → Power what-if.

Choose the proposed power parent, cable, added length, current and minimum voltage. Read the trial's displayed **Predicted end voltage**, **Predicted voltage headroom** and status before applying it. The preview recalculates the copied circuit internally; it does not display total trial supply current. This calculation uses a copied circuit until you choose **Keep in power circuit**.

Keep adds the proposed point to the working power plan. Open **Power → Load** to review the revised total supply current/loading and all section results, then export the revised job. Resetting the training exercise restores its baseline after preserving the edited copy; for real work, maintain a deliberately saved/exported baseline before keeping trials.

Complete's trial adds **300 ft P3 .625 and 0.75 A after GaN A2**. The independent reference answer is **2.798 A** supply current, **18.65%** loading and **87.37 V** at the new load. These are planning estimates for this topology, not a general cable/device allowance.

ONX files, original evidence and comparisons

Where: Job → ONX files. **Practice:** Complete → Synthetic meter files; backup step.

Function	Inputs and use	Result and limit
Open project backup	Select a full NightSweep project JSON	Opens an entire saved job. To add evidence to the current job, use the file-attachment action instead
Evidence classification	Choose Sweep, Ingress, OneCheck or supported attachment context; stage Before, After or Reference; asset/station and notes	Labels the original file and makes before/after comparisons interpretable
Add meter files or photos	Choose available files and wait for completion	Stores original bytes with the job; does not control a meter or read an image's numbers automatically
Supported CSV preview	Supply explicit frequency and level/response/channel columns	Decodes supported interchange data; proprietary ONX JSON/XML/ZIP/PDF and other native schemas remain originals unless supported
View decoded CSV measurements	Expand the parsed trace/channel table	Inspect values, units and source context; previewing applies no silent TP correction

Function	Inputs and use	Result and limit
Sweep / ingress before-after	Match compatible asset, kind, units and frequencies/stages	Compares entered file data; does not invent values or operator acceptance thresholds
OneCheck/channel table	Review supported channel, frequency, level and MER columns	Preserves discrete channel data; SC-QAM and OFDM PLC/block quantities are not interchangeable
Original-only attachments	Retain unsupported formats or photos	Preserves evidence without pretending it has measurement decoding or OCR
Save original	Export the exact stored original	Finish the device save action and verify the destination file
Verify saved files	Check current-job originals for missing/tampered bytes	Verifies stored file integrity, not a separate external backup or physical measurement
Remove from job	Remove the intended attachment reference	Protect/export evidence before removing it; removal is not a backup
Clear unused stored files	Clean originals no longer referenced by working/named jobs	Validates references first and stops conservatively on corrupt named-job data; do not use storage clearing as fault recovery

Supported interchange examples:

```
frequency_mhz,level_dbmv
54,31
860,43
```

Use `response_db` for relative dB response; a channel table can use `channel,frequency_mhz,level_dbmv,mer_db`. Provide complete finite values, explicit units and valid unique frequency/channel rows. A filename in a record is only a reference; attach the bytes if the original must travel with the backup.

Evidence/project limit	Maximum
One original attachment	8 MiB
Total original evidence in one job	12 MiB
Attachments in one job	30
Text decoded for a preview	2 MiB
CSV data rows for decoding	10,000

Evidence/project limit	Maximum
Full project JSON import/export	20 MiB

An oversized/unsupported preview can remain an original attachment within the original-file limits. Every project-opening route applies the common **20 MiB UTF-8 text limit before parsing or changing the working job**. Oversized or non-text input is rejected. A missing or tampered original blocks a portable export rather than silently dropping evidence. Wait for imports/exports to finish before switching jobs.

Field checklists and unit conversion

Where: Job → Checklists. **Practice:** Complete → Field checklists.

Function	Inputs and use	Output and limit
Leakage / ingress quick reference	Review shielding, fittings, unused ports, drops/premises and retest guidance	Definitions and inspection prompts, not leakage thresholds or automatic fault classification
Leakage repair checklist	Mark fittings, shielding/housings, terminations, affected drop/premises and retest items actually reviewed/performed	Saved work checklist; not RF acceptance
Leakage/ingress notes	Enter location, findings, repair, retest and trace references	Included with the job and relevant reports
Preventive maintenance / EOL	Record supply inspection, bonding, EOL points, amplifier/unity-gain and patrol notes	Individual checkmarks plus findings; route predictions are not copied into measured notes
dBµV ↔ dBmV	Enter the voltage level and starting unit	dBmV = dBµV − 60; reverse adds 60. Same reference plane; no TP addback or µV/m field-strength conversion

Practice **72 dBµV =12 dBmV**. This is a voltage-reference conversion, not dBm power, leakage field strength or proof of a measured level.

Project records and closeout

Where: Job → Records. **Practice:** Complete → Records; DSIM → findings; focused equipment guides.

Identity, profiles and acceptance criteria

Record ticket/node/design revision, lead, meter/calibration, reference ID, reviewer, acceptance date, disposition and notes. Job ticket/node/lead are linked with the corresponding Details fields.

Select the actual profile and enter **installed equipment upper**, **planned upper**, **available spectrum**, return ceiling/forward floor, forward/return marker frequencies, design or receiver targets, level/tilt/ripple tolerances, minimum MER, network injection targets/tolerance, control/procedure and marker basis. Mark approval only when the operator's criteria are genuinely approved.

A profile's name/container is not proof of its entered operating limit. For Complete, the profile container is labelled 1002 while the actual entered hardware/planning upper is **1000 MHz** and live upper **860 MHz**. Forward comparisons check their hardware, planned/live limits and valid ordered markers. Return comparisons also reject overlapping return/forward passbands. Resolve the reported band issue before interpreting a within-limits result.

Forward measurements

Use **Add** for a separate station/port/stage record. Enter date, node/station, profile, Before/After stage, **Planning or Measured** basis, measurement plane, actual low/high markers, raw readings, remaining addbacks, targets, ripple, trace/reference, trace-saved status, reviewer and notes. Forward rows also record model, MER, input pad/EQ, interstage trim and control system.

Read corrected low/high levels, tilt, target errors, tilt error, coverage and status. Planning data or unapproved/incomplete criteria cannot establish acceptance. A few passing markers do not certify the full trace, all frequencies or physical alignment.

Return measurements and injection setting

Return rows are for an **injected return test with a verified absolute receiver plane**. Enter the common station/stage/plane data plus injected low/high network levels, existing pad, ingress check, EQ and injection source. Referenced DOCSIS return sweepless belongs in Sweep setup, in matching dB units; do not disguise it as an absolute receiver record.

The estimated pad follows the entered receiver error and existing pad. A negative estimate calls for gain/path review; it is not silently changed to zero. Keep the separate **38 dBmV return TX reference** distinct from the chosen receiver targets and network injection levels.

The Records **Return injection setting** helper uses desired network-plane injection plus **remaining uncompensated coupling and lead** for a manual-setting estimate. Practice **40 + 20 + 1 = 61 dBmV** as form arithmetic only. This does not establish an ONX limit, physical output with compensation enabled, or a setting for the 42 MHz Return Sweepless Sweep method. The 204 Sweep setup physical-output budget uses total physical losses and has a different purpose.

Repairs, DSIM and record management

Function	Inputs and use	Output and limit
Repairs / punch list	Item, station/port, category, finding, action, owner, due/repair dates, retest, status and reviewer	Closed without required repair/retest/reviewer information remains incomplete
GainMaker / ACI DSIM record	Exact station/module/revision, plant/route, modes, pilot/type/lock, fallback, temperature, S1/pad/backoff/TGC, final verification and technician confirmation	Checks completeness/consistency and actual calendar dates for the supported DSIM-SG review; impossible imported dates remain invalid without rewriting the saved value. This is not a universal GaN/AOI/DSIM-A setup procedure
Add / remove record	Add a row or remove the intended record using the controls shown	Enter and review the actual identity, stage and measurement plane for each row
Suggest trace filename	Use the button on a forward/return row to generate a suggested name	Fills the trace reference and sets trace-saved status to **No**. It does not create, save or attach a meter file
Export section CSV	Export forward, return, punch or DSIM rows with their computed/status fields	Section report only, not the full project or original evidence
Summary and disposition	Read total records, After rows within entered limits and open/incomplete repairs	Reviewer disposition remains an explicit decision; no software calculation certifies the plant

The synthetic DSIM form in Complete is a separate form exercise. Use the dedicated DSIM job for its matching setup path and preserve the exact revision conflict. A completed setup record remains separate from forward/return acceptance.

Run sheet, exports and full backups

Where: Job → Run sheet; Files menu; relevant calculator/record task. **Practice:** Complete → Run sheet and exports / Backup.

Run sheet combines job/work-order context, selected frequency, source/model/split/limits, RF predictions, entered meter comparisons, as-built notes, field work, equipment/sweep setup and available-carrier estimates. Review its synthetic/sample warnings and missing values before sharing.

Output/function	Where and use	What it preserves
Print run sheet	Run sheet → Print	Printable field summary/PDF via device print service; actual pagination is device-dependent
Share	Run sheet → Share	Text through Android/browser share where supported, or the available prepared-file/download fallback; choose a recipient/app yourself
Export sheet CSV	Run sheet	Reviewable summary with context; not a full backup
Export RF levels	Files	Route predictions at listed frequencies; unavailable results blank
Export frequency results	Test → Frequency	Single-path inputs, frequencies, units, flags and results
Export power load	Power → Load	Power topology, loads, per-section results and qualifications
Record CSVs	Job → Records	The selected section's entered rows and computed review fields
Export setup notes	Test → Sweep setup	Setup drafts, guide/context and saved station snapshots
Save original	Job → ONX files	Exact original attachment bytes
Save project / Home backup	Files or Home	Full supported route, power, job, records, field/setup drafts, training state and original evidence
Open project / recover backup	Files, Home or ONX project shortcut	Validates and restores the entire supported project; verify route and originals after opening

Your file is ready means bytes are prepared. Choose **Save/share file** or **Download file** when offered, complete the phone's final save action, then confirm the result in Files/Downloads or the chosen destination. Cancellation is not a completed export. A PDF, screenshot, CSV, named job or prepared-file message is not a full external backup.

For a recovery rehearsal: export Complete, verify the JSON exists outside NightSweep, reopen it, confirm the job, route, power, records, practice progress and all six originals, then return to the protected real job. Keep independent backups before updates, uninstalling, clearing storage or moving among app/browser stores.

Offline use, updates and Help

Where: browser Home → NightSweep on this device; header/Home → Help. **Practice:** Complete for backup; Custom curves → Help; any loaded exercise for an offline calculation check.

Function	Use	Result and limit
Ready for offline use	Wait for the installed browser shell to report readiness	Confirms required cached shell availability; a visible online page alone is insufficient
Install app	Use the browser's offered prompt where supported	Optional Home Screen installation; full phone-specific steps are in the separate installation guide
Retry offline setup	Retry the reported failed shell preparation while online	Does not require clearing job storage
Check for updates	Look for a newer hosted release	Does not automatically replace the running app
Install update → Reload app	Export current work, apply the offered update, then reload when ready	User-controlled update sequence; check version and saved jobs afterward
Protect local storage	Request persistence where supported	Best effort, not a backup or protection from device/browser clearing/uninstall
Keep your work safe	Review local storage guidance/status	Helps identify the active store; no cloud sync or cross-browser sharing is added
Help task shortcuts	Choose the matching task from Help	Opens the existing live screen
Methods and limits	Read equations, assumptions, terminology and unavailable-data behavior	Explains interpretation; source links need connectivity
Run build self-check	Run bundled reference cases and read status	Checks those build cases; a pass is not physical device or plant certification. A failure warns against trusting results

Once ready, the installed browser app can reopen its cached shell and use bundled calculations/training offline. External manuals still need internet. The standalone HTML and native wrapper have different delivery/storage behavior; opening a local HTML file does not establish browser PWA installation or reliable local-file storage.

Detailed Android/iOS installation, compatible Preview update precautions and APK verification are documented in the separate installation guide.

Checked example answers

These are reference answers for the unchanged **synthetic** examples. Restore a baseline before comparing. They do not establish field acceptance or manufacturer settings.

Exercise/function	Inputs	Checked answer
Complete launch	54/860 MHz, raw 11/23 dBmV, remaining TP 20 dB	31/43 dBmV
Gain helper	54 MHz, input 7.1/output 16.1 dBmV, +20 dB each	Corrected 27.1/36.1 dBmV; net gain 9 dB
Complete A1 output	Entered route and 9/18 dB gain anchors	About 36.94/47.61 dBmV at 54/860 MHz
Complete A2 output	Entered route and 54/550/860 MHz gain table	About 40.55/45.62 dBmV at 54/860 MHz
Complete EOL Tap C	Entered mixed-brand route	About 14.01/4.94 dBmV at 54/860 MHz
Complete drop loss	75 ft custom RG-6, rates 1.8/6.2 dB per 100 ft plus custom 4 dB	Cable 1.35/4.65 dB; total 5.35/8.65 dB at 54/860 MHz
Complete TDR	A2 origin 1700 ft; displayed 295 ft; lead 25 ft; equal VF 0.87	Path 270 ft; coordinate 1970 ft; 160 ft into r5's 310 ft span
Complete frequency checker	860 MHz; 200 ft × 2.2 dB/100 ft + 1.4 dB; 43 dBmV launch	Loss 5.8 dB; receive 37.2 dBmV
Complete power	90 V / 15 A; synthetic 0.75/0.60/0.45 A loads; 20 W / PF 0.9 branch	2.047 A; 13.65% loading; minimum 88.83 V at A2
Complete power trial	Add 300 ft P3 .625 and 0.75 A after A2	2.798 A; 18.65%; 87.37 V at new load
Six serial AOI actives	Each incoming 300 ft custom span loses 3/9 dB; each entered net gain restores 3/9 dB	All six marker outputs 34/46 dBmV at 261/993 MHz; source node excluded from six-active count
Custom reverse TDR	100 ft displayed, actual VF 0.88/meter VF 0.80, lead 25 ft; reverse from 200 ft	Path 85 ft; route coordinate 115 ft
Voltage unit converter	72 dBµV	12 dBmV, same plane
Separate Records injection arithmetic	Desired 40 + remaining coupling 20 + remaining lead 1	61 dBmV manual-setting estimate; not automatically ONX Injection

The fresh independent complete-workflow audit compares actual application values with Wolfram references, including baseline/trial power and the explicit custom drop. Retained evidence lives with the release under `research/alpha18/`. Arithmetic agreement checks the supplied assumptions and inputs; it does not verify installed equipment, waveform, actual losses or phone behavior.

37-feature coverage index

This index follows the feature choices in the app. Choose the same label in Home's feature index to load the matching seeded practice. Sections above also cover controls that share one indexed feature, such as export buttons and station history.

#	Feature choice	Reference section	Loadable exercise
1	Home, practice and backup	Header and Home	Complete
2	Help, methods and build self-check	Offline and Help	Custom curves
3	Launch, source, correction and temperature	RF setup	Complete
4	Route, branches and distances	Cable route	Complete
5	Amplifier gain and reserve	Gain	Complete
6	Installed equipment and exact manuals	Equipment setup	Complete
7	Equipment catalog and source references	Catalog	Complete
8	Forward and return setup	Sweep setup	Complete
9	Forward TDR and active boundaries	TDR	Complete
10	Absolute, normalized and loss charts	Predicted sweep	Complete
11	Single-path frequency calculator	Frequency	Complete
12	Meter readings and prediction error	Readings	Complete
13	Cable, tap and target references	References	Complete
14	Power load, voltage and branches	Power	Complete
15	Power what-if and Keep	What-if	Complete
16	Named jobs and work-order details	Job details	Complete
17	Original files and supported CSV	Evidence	Complete
18	Leakage, maintenance and conversion	Checklists	Complete
19	Forward, return, punch and review	Records	Complete
20	Run sheet, share, print and CSV exports	Exports	Complete
21	Six serial active devices	Practice library, checked answers	Six actives

#	Feature choice	Reference section	Loadable exercise
22	Entered amplifier gain table	Gain	Six actives
23	Derive gain from two corrected readings	Gain helper	Complete
24	Tap faces and RG-6 drop splitter	Point tools	Complete; explicit custom loss
25	Map note and step-by-step arithmetic	Point tools	Custom curves
26	Custom cable and passive loss tables	Cable route	Custom curves
27	Manual source and launch table	RF setup	Custom curves
28	Reverse TDR and velocity correction	TDR	Custom curves
29	Available carriers and local targets	References	DSIM
30	OEM AGC installed control path	Equipment controls	OEM AGC
31	DSIM and historical pilot-code lookup	Equipment controls	DSIM
32	Manual / thermal installed control path	Equipment controls	Manual / thermal
33	Conventional GS7K procedure	Equipment controls	GS7K node
34	dBµV ↔ dBmV conversion	Unit conversion	Complete
35	Return injection and receiver records	Records, sweep setup	Complete
36	Before / after evidence comparison	Evidence	Complete
37	Portable backup and recovery	Full backups	Complete

Known limits and unresolved information

Exact installed **GainMaker GaN control/commissioning**, **Quantum18 LE/BA I&O**, **EN9000/RFAM/RPD commissioning**, and some **DSIM-A/module-specific** procedures remain documentation gaps. Preserve them as unresolved; another model's manual is not a substitute. Original manuals, actual firmware/options, approved pilots, physical measurement planes, actual available carriers and operator criteria require job-specific confirmation.

NightSweep does not connect to, program or configure ONX, nodes, amplifiers, DSIM or XPERTrak. Unsupported meter formats remain originals; no OCR or automatic reading extraction is added. No source table, synthetic file, checklist, software status or reference calculation establishes installed RF acceptance.

Automated checks, independent arithmetic and desktop mobile-viewport testing do not establish physical Android/iPhone installation, WebView/Safari retention, native save/share/print behavior, touch/rotation, storage eviction or actual quota exhaustion. Preserve external backups and separately verify the phone and plant used for field work.